

Def. Let X, Y be Metric Spaces, and $f_n: X \rightarrow Y$ be function defined for each $n \in \mathbb{N}$.
For each $x \in X$, define $f(x) \stackrel{\text{def}}{=} \lim_{n \rightarrow \infty} f_n(x)$ if the limit exists.

This f is called limit function of $\{f_n\}$.

Def. Suppose that a sequence of functions $\{f_n\}$ on X into Y satisfies that:
the limit function f is defined on X and

For all $\epsilon > 0$, there exists $N \in \mathbb{N}$ s.t. $n \geq N \Rightarrow d_Y(f_n(x), f(x)) \leq \epsilon, \forall x \in X$.

Then, we say that $\{f_n\}$ converges uniformly on X to f . Denote $f_n \rightrightarrows f$.

Note: $\forall x \in X, d_Y(f_n(x), f(x)) \leq \epsilon \Leftrightarrow \sup_{x \in X} d_Y(f_n(x), f(x)) \leq \epsilon$. That is,

$$f_n \rightrightarrows f \text{ on } X \Leftrightarrow \lim_{n \rightarrow \infty} \sup_{x \in X} d_Y(f_n(x), f(x)) = 0.$$

Thm. Let $\{f_n\}$ be functions defined on any Metric Space X into Complete Metric space Y . Then,
In General Metric Space

$\{f_n\}$ converges uniformly on X to f if and only if $\forall \epsilon > 0, \exists N \in \mathbb{N}$ s.t. $m, n \geq N \Rightarrow \sup_{x \in X} d_Y(f_n(x), f_m(x)) \leq \epsilon$

Proof. If $f_n \rightrightarrows f$, then, In Complete Metric space

$$\sup_{x \in X} d_Y(f_n(x), f_m(x)) \leq \sup_{x \in X} d_Y(f_n(x), f(x)) + \sup_{x \in X} d_Y(f(x), f_m(x)) \rightarrow 0$$

Give the result. (간단하게 보려면, Supremum의 성질과 triangle inequality가 활용되면 풀 수 있음.)

Conversely, since Cauchy sequence in the Complete Metric space converges, the limit function f exists.
Given $\epsilon > 0$, choose $N \in \mathbb{N}$ such that the Cauchy condition holds.

For fixed $n \geq N$, for all $x \in X$ and $m \geq N$, $d_Y(f_n(x), f_m(x)) \leq \epsilon$.

So, letting $m \rightarrow \infty$, we obtain $d_Y(f_n(x), f(x)) \leq \epsilon$, for all $x \in X$. Thus proved. $\square$

Fail Try.

$$\lim_{m \rightarrow \infty} \sup_{x \in X} d_Y(f_m(x), f_n(x)) = \sup_{x \in X} d_Y(f(x), f_n(x))$$

Gives the result. (More specifically, let $n \in \mathbb{N}$ fixed. Observe that

$$\left| \sup_{x \in X} d_Y(f_m(x), f_n(x)) - \sup_{x \in X} d_Y(f(x), f_n(x)) \right| \xrightarrow{d_Y \text{ continuous}}$$

(*)

$$\sup_{x \in X} d_Y(f(x), f_n(x)) = \sup_{x \in X} d_Y(\lim_{m \rightarrow \infty} f_m(x), f_n(x)) = \sup_{x \in X} \lim_{m \rightarrow \infty} d_Y(f_m(x), f_n(x)) = \lim_{m \rightarrow \infty} \sup_{x \in X} d_Y(f_m(x), f_n(x))$$

$$\left| \sup_{x \in X} \lim_{m \rightarrow \infty} d_Y(f_m(x), f_n(x)) - \sup_{x \in X} d_Y(f_m(x), f_n(x)) \right| \leq \sup_{x \in X} \left| \lim_{m \rightarrow \infty} d_Y(f_m(x), f_n(x)) - d_Y(f_m(x), f_n(x)) \right| \rightarrow 0$$

Gives (*).

Uniform Convergence and Continuity.

Thm. Suppose that $f_n, f: X \rightarrow Y$ are functions where X is Metric, Y is complete Metric sp.

Let $x \in X$ be a limit point of X .

If for each $n \in \mathbb{N}$, the limit $\lim_{t \rightarrow x} f_n(t)$ exists and $f_n \rightarrow f$ uniformly on X ,

then $\lim_{t \rightarrow x} \lim_{n \rightarrow \infty} f_n(t) = \lim_{n \rightarrow \infty} \lim_{t \rightarrow x} f_n(t)$

Proof. For each $k \in \mathbb{N}$, take $A_k = \lim_{t \rightarrow x} f_k(t)$. And, since the uniform convergence, given $\epsilon > 0$, there exists $N \in \mathbb{N}$ such that

$$m, n \geq N \Rightarrow d_Y(f_n(t), f_m(t)) \leq \epsilon \text{ for all } t \in X.$$

And, for such $m, n \geq N$, let $x_0 \in X$ such that

$$d_Y(A_m, f_m(x_0)) < \epsilon \text{ and } d_Y(A_n, f_n(x_0)) < \epsilon$$

(for some $\delta, \delta > 0$, $t \in B_\delta(x) \Rightarrow d_Y(f_k(t), A_k) < \epsilon$, by def. of A_k)

Now,

$$d_Y(A_m, A_n) \leq d_Y(A_m, f_m(x_0)) + d_Y(f_m(x_0), f_n(x_0)) + d_Y(f_n(x_0), A_n) \leq 3\epsilon,$$

Therefore $\{A_k\}$ is a Cauchy sequence in Y , so it is convergent.

Let $A = \lim_{n \rightarrow \infty} A_n$. Then,

$$d_Y(f(t), A) \leq \underbrace{d_Y(f(t), f_n(t))}_{f_n \rightarrow f \text{ uniformly}} + \underbrace{d_Y(f_n(t), A_n)}_{f_n(t) \rightarrow A_n \text{ as } t \rightarrow x} + \underbrace{d_Y(A_n, A)}_{A_n \rightarrow A} \rightarrow 0$$

Rigorously, given $\epsilon > 0$, let $N_1, N_2 \in \mathbb{N}$ s.t.

$$n \geq N_1 \Rightarrow d_Y(A_n, A) < \epsilon \text{ and } n \geq N_2 \Rightarrow \forall t \in X, d_Y(f_n(t), f(t)) < \epsilon.$$

Now, take $n \geq \max\{N_1, N_2\}$ and $\delta > 0$ s.t.

$$t \in B_\delta(x) \Rightarrow d_Y(f_n(t), A_n) < \epsilon. \quad \blacksquare$$

Corollary. If $\{f_n: X \rightarrow Y\}$ is a sequence of continuous maps on a Metric space X into Complete Y , and $f_n \rightarrow f$, then f is continuous.

Uniform Convergence with Continuity

Thm. Suppose that f_n ($n \in \mathbb{N}$) are functions on a Metric X into Complete Metric Y . Let $x \in X$ be a limit point of X .

If for each $n \in \mathbb{N}$, the limit $\lim_{t \rightarrow x} f_n(t)$ exists and f_n converges uniformly on X , then
$$\lim_{t \rightarrow x} \lim_{n \rightarrow \infty} f_n(t) = \lim_{n \rightarrow \infty} \lim_{t \rightarrow x} f_n(t).$$

Proof. By the Uniformly Convergence, given $\epsilon > 0$, there exists $N \in \mathbb{N}$ s.t.

$$n, m \geq N \Rightarrow d_Y(f_m(t), f_n(t)) \leq \epsilon \text{ for any } t \in X.$$

Thm. Let K be a Compact space, and

$f_n: K \rightarrow \mathbb{R}$ be continuous functions which converges pointwise to a continuous function f on K .

If $f_n \geq f_{n+1}$ for all $n \in \mathbb{N}$, then $f_n \rightarrow f$ uniformly.

Proof. Put $g_n = f_n - f$. Then, g_n is continuous and $g_n \geq 0$ pointwise, and $g_n \geq g_{n+1}$.

Given $\epsilon > 0$, let $K_n = \{x \in K \mid g_n(x) \geq \epsilon\} = g_n^{-1}([\epsilon, \infty))$

Since g_n is continuous, K_n is a closed set in a Compact space, so it is compact.

And, since $g_n \geq g_{n+1}$, $K_n \supseteq K_{n+1}$ (if $g_{n+1}(x) \geq \epsilon$, then $g_n(x) \geq g_{n+1}(x) \geq \epsilon$)

Fix $x \in X$. Since $g_n(x) \rightarrow 0$ as $n \rightarrow \infty$, for large enough $n \in \mathbb{N}$, $x \notin K_n$.

That is, $x \notin \bigcap K_n$, so $\bigcap K_n$ is empty. But, since each K_n is compact, thus $K_N = \emptyset$ for some $N \in \mathbb{N}$. That is,

$$0 \leq g_N(x) < \epsilon \text{ for all } x \in K,$$

and since $g_N \geq g_{N+1} \geq g_{N+2} \geq \dots$, so $\sup |g_n - 0| \rightarrow 0$. $\square$

Thm. Let X be a Metric space, and $\mathcal{C}_b(X)$ be the all complex-valued continuous, bounded functions on X .

Define $\|f\| = \sup_{x \in X} \|f(x)\|$, then it is norm on $\mathcal{C}_b(X)$. This norm can derive a Metric on $\mathcal{C}_b(X)$.

Uniform Convergence and Integration.

Thm. Let $\alpha: [a, b] \rightarrow \mathbb{R}$ be monotone increasing and $f_n \in \mathcal{B}(\alpha)$, $n \in \mathbb{N}$.

If $f_n \rightarrow f$ on $[a, b]$, then $f \in \mathcal{B}(\alpha)$ and

$$\lim_{n \rightarrow \infty} \int_a^b f_n d\alpha = \int_a^b \lim_{n \rightarrow \infty} f_n d\alpha = \int_a^b f d\alpha.$$

Proof. For each $n \in \mathbb{N}$, define $E_n = \sup_{x \in [a, b]} |f_n(x) - f(x)|$. Then, clearly

$$f_n - E_n \leq f \leq f_n + E_n \quad (\text{for all } t \in [a, b])$$

$$\text{Therefore, } \int_a^b f_n - E_n d\alpha \leq \int_a^b f d\alpha \leq \int_a^b f d\alpha \leq \int_a^b f_n + E_n d\alpha \quad (*)$$

$$\text{Hence } 0 \leq \int_a^b f d\alpha - \int_a^b f d\alpha \leq 2E_n \cdot [\alpha(b) - \alpha(a)] \rightarrow 0 \text{ as } n \rightarrow \infty.$$

Thus $f \in \mathcal{B}(\alpha)$. The equality is obtained by (*), using $f \in \mathcal{B}(\alpha)$.

Uniform Convergence and Differentiation.

Thm. Let $f_n: [a, b] \rightarrow \mathbb{R}$ be differentiable maps s.t. for some $x_0 \in [a, b]$, $\{f(x_0)\}$ converges.

If $\{f_n\}$ converges uniformly on $[a, b]$, then $\{f_n\}$ converges uniformly on $[a, b]$ to f with

$$\frac{d}{dx} \lim_{n \rightarrow \infty} f_n(x) = \lim_{n \rightarrow \infty} \frac{d}{dx} f_n(x).$$

Proof. Given $\varepsilon > 0$, take $N \in \mathbb{N}$ s.t. $n, m \geq N$ implies

$$|f_n(x_0) - f_m(x_0)| \leq \frac{\varepsilon}{2} \text{ and } |f'_n(t) - f'_m(t)| \leq \frac{\varepsilon}{2(b-a)}, \quad \forall t \in [a, b].$$

By the Mean-Value Theorem for $f_n - f_m$, for all $x, t \in [a, b]$,

$$|f_n(x) - f_m(x) - (f_n(t) - f_m(t))| \leq \frac{\varepsilon}{2(b-a)} \cdot (b-a) = \frac{\varepsilon}{2}. \quad (*)$$

Therefore, for any $x, t \in [a, b]$ and $n, m \geq N$,

$$|f_n(x) - f_m(x)| \leq |f_n(x) - f_m(x) - f_n(x_0) + f_m(x_0)| + |f_n(x_0) - f_m(x_0)| \leq \frac{\varepsilon}{2} + \frac{\varepsilon}{2} = \varepsilon.$$

so that $\{f_n\}$ converges uniformly, let $f = \lim_{n \rightarrow \infty} f_n$. For each $x \in [a, b]$, define on $[a, b] \setminus \{x\}$

$$\phi_n(t) = \frac{f_n(t) - f_n(x)}{t - x} \quad \text{and} \quad \phi(t) = \frac{f(t) - f(x)}{t - x}$$

(*) gives ϕ_n is Cauchy sequence, thus converges uniformly, to ϕ . Finally,

the fact that ϕ_n continuous gives the result.

Summary. Let $f_n: X \rightarrow Y$ be functions, $f_n \rightarrow f$.

Continuity If f_n are continuous and $f_n \rightarrow f$, then f is continuous.

Differentiation If f_n are differentiable and $f_n \rightarrow f'$ and $f_n(x_0) \rightarrow f(x_0)$, then $f_n \rightarrow f$ and

$$\frac{d}{dx} \lim_{n \rightarrow \infty} f_n(x) = \lim_{n \rightarrow \infty} \frac{d}{dx} f_n(x)$$

Integral If $f_n \in \mathcal{B}(a)$ and $f_n \rightarrow f$, then $f \in \mathcal{B}(a)$ and

$$\int_a^b \lim_{n \rightarrow \infty} f_n \, dx = \lim_{n \rightarrow \infty} \int_a^b f_n \, dx$$

Counterexamples

$$\begin{aligned} & \Gamma f_n: [0,1] \rightarrow \mathbb{R}; x \mapsto x^n & f_n \text{ conti,} \\ & \lim_{n \rightarrow \infty} f_n(x) = \begin{cases} 0 & \text{if } 0 \leq x < 1 \\ 1 & \text{if } x = 1. \end{cases} & f \text{ not conti} \end{aligned}$$

$$\begin{aligned} & \Gamma f_n: [0,1] \rightarrow \mathbb{R}; x \mapsto \frac{1}{n}x & f_n \text{ converges uniformly} \\ & \overline{f_n}: \mathbb{R} \rightarrow \mathbb{R}; x \mapsto \frac{1}{n}x & \overline{f_n} \text{ not converges uniformly} \end{aligned}$$

$$\begin{aligned} & \Gamma f_n: [0,2] \rightarrow \mathbb{R}; x \mapsto \frac{1-x}{1+x^n} & f_n \text{ differentiable, converges uniformly} \\ & \lim_{n \rightarrow \infty} f_n(x) = \begin{cases} 1-x & \text{if } 0 \leq x \leq 1 \\ 0 & \text{if } 1 < x \leq 2. \end{cases} & f \text{ not but differentiable at } x=1 \end{aligned}$$

$$\text{In } [0,1], |f_n(x) - f(x)| = \left| (1-x) \cdot \left(\frac{1}{1+x^n} - 1 \right) \right| = \frac{|1-x| \cdot |x^n|}{|1+x^n|} \leq |1-x| \cdot |x^n| = (1-x) \cdot x^n.$$

$$\begin{aligned} & \text{Since } ((1-x) \cdot x^n)' = -x^n + (1-x)n \cdot x^{n-1} = nx^{n-1} - (n+1)x^n = x^{n-1}(n - (n+1)x) \\ & \text{has a maximum at } x = \frac{n}{n+1}, \quad \left(1 - \frac{n}{n+1} \right) \cdot \left(\frac{n}{n+1} \right)^n = \frac{n^n}{(n+1)^{n+1}} \rightarrow 0 \text{ as } n \rightarrow \infty. \end{aligned}$$

$$\text{So, } \sup_{[0,1]} |f_n(x) - f(x)| \rightarrow 0 \text{ as } n \rightarrow \infty.$$

$$\text{In } [1,2], |f_n(x) - f(x)| = \left| \frac{1-x}{1+x^n} \right| = \frac{x-1}{1+x^n} \leq \frac{x-1}{x^n} = \frac{1}{x^n} - \frac{1}{x^n} \rightarrow 0 \text{ as } n \rightarrow \infty.$$

Now, $f_n \rightarrow f$ converges uniformly.

$$\begin{aligned} & \Gamma f_n: \mathbb{R} \rightarrow \mathbb{R}; x \mapsto \frac{1}{\sqrt{n}} \cdot \sin(nx). & f_n \text{ differentiable, converges uniformly} \\ & f = 0, \text{ since } |\sin(mx)| \leq 1 & \text{but} \\ & f'_n: \mathbb{R} \rightarrow \mathbb{R}; x \mapsto \sqrt{n} \cdot \cos(nx) & f'_n \text{ not converges pointwise.} \end{aligned}$$

$$\Gamma \text{ Let } X_n = \{x_n | n \in \mathbb{N}\} = \mathbb{Q} \cap [0, 1].$$

$$f_n: [0, 1] \rightarrow \mathbb{R}; x \mapsto \begin{cases} 1 & \text{if } x \in X_n \\ 0 & \text{if } x \notin X_n. \end{cases} \quad f_n \text{ Riemann-integrable, since it has finite discords.}$$

$$f: [0, 1] \rightarrow \mathbb{R}; x \mapsto \begin{cases} 1 & \text{if } x \in \mathbb{Q} \\ 0 & \text{if } x \notin \mathbb{Q} \end{cases} \quad f \text{ not locally integrable, it is Dirichlet.}$$

$$\Gamma f_n: [0, 1] \rightarrow \mathbb{R}; x \mapsto \begin{cases} n & \text{if } 0 < x < \frac{1}{n} \\ 0 & \text{if } x=0 \text{ or } \frac{1}{n} \leq x \leq 1. \end{cases}$$

$$\lim_{n \rightarrow \infty} f_n \equiv 0, \text{ but } \int_0^1 f_n = 1. \quad \text{Cannot change limit under the integral.}$$

$$\Gamma f_n: \mathbb{R} \rightarrow \mathbb{R}; x \mapsto \frac{x^2}{(1+x^2)^n} \quad f_n \text{ continuous, but the sum is not continuous.}$$

$$f(x) = \sum_{n=0}^{\infty} f_n(x) = \sum_{n=0}^{\infty} \frac{x^2}{(1+x^2)^n} = \begin{cases} 0 & \text{if } x=0 \\ 1+x^2 & \text{if } x \neq 0 \end{cases} \quad \left(x^2 \cdot \frac{1}{1 - \frac{1}{1+x^2}} \right)$$

$$\Gamma f_n: [0, 1] \rightarrow \mathbb{R}; x \mapsto nx \cdot (1-x^2)^n \quad f_n \text{ differentiable,}$$

$$\lim_{n \rightarrow \infty} f_n(x) = 0$$

$$\int_0^1 x(1-x^2)^n dx = \int_1^0 -\frac{1}{2} \cdot t^n dt = \frac{1}{2} \cdot \frac{1}{n+1} t^{n+1} \Big|_0^1 = \frac{1}{2} \cdot \frac{1}{n+1}.$$

$$\text{So, } \int_0^1 f_n(x) dx = \frac{1}{2} \cdot \frac{n}{n+1} \rightarrow \frac{1}{2}, \text{ but } \int_0^1 f dx = 0.$$

$$\text{Note: replace } nx \text{ to } n^2 x, \text{ then } \int_0^1 f_n \rightarrow \infty.$$